Network Design Framework for Small to Micro Businesses

Leadership Review Draft
March 2026 pricing snapshot

Document purpose
Provide a reproducible, low-cost network standard for 1–20 user sites with mandatory VPN, mandatory guest wireless, and a mandatory local mini server for monitoring and file storage. This document focuses on design approval, hardware selection, cost estimation, and exception framing—not configuration.

Built from the uploaded governance set plus current March 2026 product and street-price checks. Prices are estimates and may vary by seller, stock, and refurb condition.

Requirements, Goals, and Design Constraints

Governance baseline. Every approved design must support business Wi-Fi, guest isolation, IoT-capable segmentation, protective DNS, measurable uptime, monitored remote access, and backup/retention discipline.

Topic	Required outcome	Design effect
Users	1–20 users total	Two tiers are required: 1–10 and 10–20.
VPN	Mandatory remote access	Tailscale is the preferred remote-access path to avoid static-IP dependence.
Guest Wi-Fi	Mandatory at every site	Guest SSID, client isolation, internet-only policy, and traffic controls are baseline.
Server	Mandatory local mini server / PC server	Monitoring, file storage, controller software, and optional subnet routing land on the server.
Cost	As cheap as possible	Buy only what governance requires; use exceptions where needed.
Operations	Entry-level IT reproducibility	Avoid fragile one-off builds unless specifically flagged as exceptions.

Design principles

- Separate edge routing, switching, wireless, and local services unless a low-cost design deliberately collapses roles to reduce spend.
- Keep guest, trusted business, management, and optional IoT traffic in distinct zones with least-privilege access between them.
- Use a local controller or management plane whenever it materially improves repeatability and troubleshooting for junior staff.
- Place monitoring and file retention on the local server rather than depending on appliance telemetry alone.
- Treat UPS and backup storage as resilience controls; they are not optional in the recommended design even when they create cost exceptions.

Design 1 — Micro Site Standard (1–10 Users)

Best balance of cost, policy fit, and repeatability for the smallest offices.

Best-in-slot pairing	Estimated total	Best fit	Primary posture
ER605 + TL-SG108PE + EAP610 + GMKtec NucBox G2 Plus + SX650U	$474–$564 with local backup drive	Single AP, light growth, owner-operated or lightly staffed offices	Strongest value design in the document; it meets the mandatory guest, VPN, and local server requirements without inflating hardware count.

Design summary

- Strongest value design in the document; it meets the mandatory guest, VPN, and local server requirements without inflating hardware count.
- Omada controller software runs free on the mandatory server, so the network stays centrally manageable without an extra controller purchase.
- PoE switch avoids injectors and keeps the install clean for entry-level support staff.

Key watchouts / exception notes

- Server cost is the main unavoidable exception: new mini servers do not realistically land under $100 with enough room for monitoring and storage.
- One AP is appropriate only when the floor plan is modest; poor RF environments should move to Design 2.
- The ER605 is good at the price but is not a premium firewall; deeper inspection and identity features remain compensating-control territory.

Inventory and cost estimate

Role	Best-in-slot device	Est. street price	Why it is here
Router	TP-Link ER605	$60	Low-cost routed edge with VLANs and controller awareness
Switch	TP-Link TL-SG108PE	$85	8 managed ports, 4 PoE+ ports, simple small-site fit
Wireless	TP-Link EAP610	$99	Business Wi-Fi 6 with multi-SSID and guest portal support
Server	GMKtec NucBox G2 Plus	$150	Monitoring, file storage, Tailscale, Omada software controller
UPS	CyberPower SX650U	$80	Basic outage and power-quality protection for the stack

Alternative devices for this design

Role	Alternative device	Est. price	Tradeoff

Router	GL.iNet Brume 2	$56–$80	Cheaper in some channels, but no Wi-Fi and less clean ecosystem alignment
Switch	TP-Link SG2008	$65–$85	Cleaner management than Easy Smart, but no PoE
Wireless	TP-Link EAP650	$97–$121	Better RF headroom, slightly higher spend
Server	Refurb Lenovo ThinkCentre Tiny	$79–$199	Can cut cost, but refurb variance adds lifecycle risk

Design 2 — Small Site Standard (10–20 Users)

The recommended growth design when leadership wants fewer near-term upgrades.

Best-in-slot pairing	Estimated total	Best fit	Primary posture
ER707-M2 + SG2008P + 2× EAP610 + GMKtec NucBox G2 Plus + SX650U	$714–$804 with local backup drive	Busier offices, more floor area, higher guest usage, and moderate growth expectations	Adds a second AP and stronger router/switch pairing so the design absorbs growth without immediate hardware replacement.

Design summary

- Adds a second AP and stronger router/switch pairing so the design absorbs growth without immediate hardware replacement.
- Controller-led switching and wireless improve repeatability for junior admins compared with mixed-vendor sprawl.
- Better fit where guest wireless is always on and the site cannot tolerate RF dead spots.

Key watchouts / exception notes

- This tier breaks the strict per-device sub-$100 target more often, especially at the router and switch tiers.
- Two APs should be justified by floor plan or capacity—not added by habit in tiny offices.
- Still not a true enterprise NGFW stack; it is a strong SMB design, not an IDS/IPS platform.

Inventory and cost estimate

Role	Best-in-slot device	Est. street price	Why it is here
Router	TP-Link ER707-M2	$100–$125	Better WAN/VPN posture and 2.5GbE headroom
Switch	TP-Link SG2008P	$90–$103	Smart PoE switch with stronger management and 62 W budget
Wireless	2 × TP-Link EAP610	$198	Two business APs for cleaner coverage and capacity
Server	GMKtec NucBox G2 Plus	$150	Same local services role as Design 1
UPS	CyberPower SX650U / similar	$80	Protects the fuller stack

Alternative devices for this design

Role	Alternative device	Est. price	Tradeoff
Router	TP-Link ER605	$60	Acceptable cost-cut if WAN and growth needs are modest

Switch	TP-Link TL-SG108PE	$85	Cheaper, but lighter feature set and lower growth comfort
Wireless	2 × EAP650	$194–$242	Better Wi-Fi headroom for denser user profiles
Server	Beelink EQ12-class mini PC	$180–$240	Potentially stronger compute, but higher cost

Design 3 — Compact All-in-One Edge Design

A lower-hardware-count design that trades ecosystem consistency for simplicity.

Best-in-slot pairing	Estimated total	Best fit	Primary posture
GL.iNet Flint 2 + SG2008P + GMKtec NucBox G2 Plus + SX650U	$485–$575 with local backup drive	Compact offices that want fewer boxes and can live without controller-managed routing	Collapses router and AP roles into one Wi-Fi 6 edge device, reducing hardware count and install complexity.

Design summary

- Collapses router and AP roles into one Wi-Fi 6 edge device, reducing hardware count and install complexity.
- Still preserves the required local server for monitoring, storage, and Tailscale services.
- Good fit when a small office wants a single prominent edge box rather than separate router and AP devices.

Key watchouts / exception notes

- This is less elegant for standardized fleet operations because wireless and routing are no longer in the same managed platform as the switch.
- Single-device Wi-Fi at the edge can be a poor fit when the ISP handoff location is not the best RF location.
- At current pricing, it is not materially cheaper than the recommended Omada micro standard.

Inventory and cost estimate

Role	Best-in-slot device	Est. street price	Why it is here
Router + Wi-Fi	GL.iNet Flint 2	$156–$160	Wi-Fi 6 all-in-one edge with modern port set
Switch	TP-Link SG2008P	$90–$103	Adds managed downstream ports and PoE headroom
Server	GMKtec NucBox G2 Plus	$150	Carries controller/monitoring/storage/Tailscale roles
UPS	CyberPower SX650U	$80	Required resilience component
Backup	External USB drive	$90	Local backup target

Alternative devices for this design

Role	Alternative device	Est. price	Tradeoff
Router + Wi-Fi	MikroTik hAP ax2	$99–$123	Cheaper, but steeper operational learning curve
Switch	Zyxel GS1200-5	$20–$26	Ultra-cheap, but only 5 ports and no PoE

Server	Refurb Lenovo ThinkCentre Tiny	$79–$199	Cheaper path with lifecycle caveat
UPS	APC BE425M	$60	Lower-capacity fallback UPS

Design 4 — Ultra-Low-Cost Exception Build (Sub-$280)

All-new hardware lineup versus the earlier designs; built to hit cost first while still satisfying mandatory functions.

Best-in-slot pairing	Estimated total	Best fit	Primary posture
MikroTik hAP ax2 + Zyxel GS1200-5 + refurb Lenovo ThinkCentre Tiny + APC BE425M	$258–$269 typical target	Very small offices, pilot sites, or budget-limited exceptions where used hardware is acceptable	This is the only design in the document intentionally engineered to stay under $280 total while still covering guest Wi-Fi, VPN pathing, and local server requirements.

Design summary

- This is the only design in the document intentionally engineered to stay under $280 total while still covering guest Wi-Fi, VPN pathing, and local server requirements.
- MikroTik provides Wi-Fi 6, VLAN-capable routing, and captive-portal capability in one box, letting the design avoid a separate AP and router purchase.
- The cheap managed Zyxel switch keeps a few extra ports available for printers, server, or admin uplinks without blowing the budget.

Key watchouts / exception notes

- This is an exception design, not the standard recommendation: it depends on used/refurb server hardware and a more complex RouterOS operational model.
- Lower-end UPS capacity means shorter runtime and less margin for additional devices.
- Port count and PoE capability are constrained, so this build should not be selected for growth-minded sites.

Inventory and cost estimate

Role	Best-in-slot device	Est. street price	Why it is here
Router + Wi-Fi	MikroTik hAP ax2	$99–$123	All-in-one Wi-Fi 6 edge with RouterOS flexibility
Switch	Zyxel GS1200-5	$20–$26	5-port managed switch with VLAN/QoS basics
Server	Refurb Lenovo ThinkCentre Tiny	$79–$99 target	Cheapest practical way to host monitoring and file storage
UPS	APC BE425M	$60	Entry UPS for constrained micro stacks

Alternative devices for this design

Role	Alternative device	Est. price	Tradeoff
Server	Refurb Dell OptiPlex Micro	$90–$130	Equivalent used-market option if Lenovo pricing is unfavorable

Switch	TP-Link TL-SG105E	$25–$35	Another low-cost managed 5-port option
Router + Wi-Fi	GL.iNet Beryl AX	$90–$110	Smaller and simpler, but not as policy-flexible as MikroTik
UPS	Refurb / open-box 450–650 VA UPS	$40–$70	Possible savings, but battery health uncertainty grows

Cross-Design Inventory Overview

Read this page as the procurement view. It compresses all designs into one inventory-first table so leadership can compare total spend, hardware count, and exception intensity before approving a standard.

Design	Router / edge	Switch	Wireless	Server + UPS	Total est.
Design 1	ER605	TL-SG108PE	1× EAP610	G2 Plus + SX650U	$474–$564
Design 2	ER707-M2	SG2008P	2× EAP610	G2 Plus + SX650U	$714–$804
Design 3	Flint 2	SG2008P	Built into edge	G2 Plus + SX650U	$485–$575
Design 4	hAP ax2	GS1200-5	Built into edge	Refurb Tiny + BE425M	$258–$269

Procurement note. The only design that cleanly lands under enterprise-like cost while still meeting all mandatory functions without a major governance compromise is the Omada micro standard. The sub-$280 build is feasible, but it is intentionally a constrained exception design with used hardware and tighter operational margins.

Recommended Standard: Design 1 — Omada Micro Site Standard

Recommended complete pairing: TP-Link ER605 + TL-SG108PE + EAP610 + GMKtec NucBox G2 Plus + CyberPower SX650U + external backup drive

Why this pairing wins

- Lowest-cost complete design that still keeps routing, switching, wireless, monitoring, and storage in the right places.
- Supports mandatory guest Wi-Fi and scales cleanly from 1–10 users without redesign.
- The free Omada software controller can run on the required local server, reducing controller appliance spend to zero.
- The design keeps Tailscale off the edge as a hard dependency; the server can handle subnet-router duties, identity-based remote access, and logging while the edge stays simple.
- This is the easiest stack for entry-level IT to reproduce and troubleshoot because all primary network components live in one ecosystem.

Recommended logical zones

Zone	Purpose	Default access posture
Trusted	Employee and business devices	Internet + approved internal services
Guest	Visitors and contractors	Internet only + client isolation + bandwidth controls
IoT	Optional smart devices / printers	Internet only unless explicitly allowed elsewhere
Management	Network administration	Infrastructure only
Server	Monitoring, storage, controller, Tailscale	Restricted inbound from Trusted/Admin only

Detailed inventory for the recommended standard

Role	Device	Est. price	Feature fit	Exception status
Edge router	TP-Link ER605	$60	VLAN, firewall policy, WAN flexibility, controller integration	No exception
Managed switch	TL-SG108PE	$85	8 ports, 4 PoE+ ports, VLANs, enough power for one AP	No exception
Access point	EAP610	$99	Business Wi-Fi 6, guest SSID, captive portal support via controller	No exception
Mini server	GMKtec NucBox G2 Plus	$150	Local monitoring, file storage,	Cost exception

			Tailscale subnet router, Omada software controller	
UPS	CyberPower SX650U	$80	Short outage ride-through and cleaner shutdown behavior	No exception
Backup media	External 2 TB USB drive	$90	Supports 3-2-1 style local backup copy	Optional but recommended

Recommended deployment note: The edge and AP stay purpose-built, while the required server absorbs controller, monitoring, backup, and Tailscale responsibilities. This is the most balanced spend profile in the document.

Device Cost-Feature Analysis Report

Final approval lens. This page scores the hardware itself so leadership can approve device classes independent of any one design. Prices are March 2026 estimates; refurb values fluctuate the most.

Device	Est. price	Best strengths	Main compromise	Best use	Value view
TP-Link ER605	$50–$70	Cheap VLAN-capable routed edge with Omada fit	Less headroom than pricier gateways	Default 1–10 user router	Excellent
TP-Link ER707-M2	$100–$125	2.5GbE and stronger WAN/VPN posture	Often exceeds strict sub-$100 target	Default 10–20 user router	Very strong
GL.iNet Flint 2	$156–$160	Cuts box count with Wi-Fi 6 all-in-one edge	Less clean for standardized controller-led fleets	Compact small offices	Good
MikroTik hAP ax2	$99–$123	Very flexible all-in-one edge with captive portal options	Operational learning curve is steeper	Budget exception build	Very good
TL-SG108PE	$80–$86	Best-value PoE managed switch for micro sites	Feature depth is lighter than SG2008P	Default micro switch	Excellent
SG2008P	$90–$103	Better manageability and growth headroom	Can cross budget ceiling depending on seller	Default small-site switch	Very strong
Zyxel GS1200-5	$20–$26	Extremely cheap managed switching	Only 5 ports and no PoE	Ultra-budget build	Excellent for price
EAP610	$97–$99	Business Wi-Fi 6 with strong SMB feature fit	One AP may not be enough beyond micro layouts	Default AP choice	Excellent
GMKtec NucBox G2 Plus	$150	Best new low-power server for controller, monitoring, storage	Cost exception versus $100 target	Recommended standard	Very strong
Refurb Lenovo ThinkCentre Tiny	$79–$199	Cheapest practical local server path	Refurb lifecycle and condition variance	Budget and exception sites	Very strong
CyberPower SX650U	$70–$85	Good right-sized UPS for micro stacks	Runtime margin is modest with growth	Recommended default UPS	Good
APC BE425M	$60	Lowest practical UPS entry point	Lower capacity and less expansion room	Ultra-budget build	Fair–Good